Yangyuanchen Liu

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Research and Teaching Experience

Postdoctoral Fellow | Oct 2024–Present *Johns Hopkins University*, Baltimore, MD

- Postdoctoral advisor: Somnath Ghosh
- Integrate scientific machine learning (neural operators) with multiscale multiphysics models for nondestructive evaluation (NDE).

Research Aide Technical | May 2023–Aug 2023 *Argonne National Lab*, Lemont, IL

- MOOSE finite element software development for high-performance computing (HPC).

Teaching Assistant | Aug 2020–May 2021 *Duke Kunshan University*, Kunshan, China

- Graded assignments and led recitation sessions for Linear Algebra and Stochastic Processes.

Education

Doctor of Philosophy (PhD) | Sep 2020–Sep 2024 *Duke University*, Durham, NC

- Mechanical Engineering, Computational Mechanics track; Advisor: John Dolbow
- Extended phase-field fracture theory explicitly incorporating material strength surfaces to dynamic loading, with large-scale simulations of crack nucleation and propagation.
- Developed models and numerical simulations for coupled acoustics, elastodynamics, and damage, and for pulsed laser ablation.
- Contributed to the development of RACCOON, an open-source finite element code for phase-field fracture.
- Dissertation: *A computational framework for simulating crack nucleation and growth in materials subjected to dynamic loads*

Master of Science (MS) | Sep 2017–Mar 2020 *Shanghai Jiao Tong University*, Shanghai, China

- Advisor: Yongxing Shen

Bachelor of Engineering (BE) | Sep 2013–Jul 2017 *Jilin University*, Changchun, China

Publications and Preprints

[1] **Liu, Y.**, & Ghosh, S. (2026). Neural operator-based damage sensing digital twin for piezocomposites with parametrically upscaled coupled constitutive damage model. *Computer Methods in Applied Mechanics and Engineering*, *461*, 119169. https://doi.org/10.1016/j.cma.2026.119169

[2] Torres, D. E., Zeng, B., Costa, A., **Liu, Y.**, Jiang, W., Dolbow, J. E., & Hu, T. (2026). RACCOON: A versatile framework for variational approaches to fracture with coupled field phenomena. *Advances in Engineering Software*, *221*, 104242. https://doi.org/10.1016/j.advengsoft.2026.104242

[3] **Liu, Y.**, Lopez-Pamies, O., & Dolbow, J. E. (2024). On the effects of material strength in dynamic fracture: A phase-field study. *arXiv Preprint*. https://doi.org/10.48550/arXiv.2411.16393

[4] **Liu, Y.**, Zhong, P., Lopez-Pamies, O., & Dolbow, J. E. (2024). A model-based simulation framework for coupled acoustics, elastodynamics, and damage with application to nano-pulse lithotripsy. *International Journal of Solids and Structures*, *289*, 112626. https://doi.org/10.1016/j.ijsolstr.2023.112626

[5] Kumar, A., **Liu, Y.**, Dolbow, J. E., & Lopez-Pamies, O. (2024). The strength of the Brazilian fracture test. *Journal of the Mechanics and Physics of Solids*, *182*, 105473. https://doi.org/10.1016/j.jmps.2023.105473

[6] **Liu, Y.**, Claus, S., Kerfriden, P., Chen, J., Zhong, P., & Dolbow, J. E. (2023). Model-based simulations of pulsed laser ablation using an embedded finite element method. *International Journal of Heat and Mass Transfer*, *204*, 123843. https://doi.org/10.1016/j.ijheatmasstransfer.2022.123843

[7] Xiang, G., Chen, J., Ho, D., Sankin, G., Zhao, X., **Liu, Y.**, Wang, K., Dolbow, J., Yao, J., & Zhong, P. (2023). Shock waves generated by toroidal bubble collapse are imperative for kidney stone dusting during Holmium:YAG laser lithotripsy. *Ultrasonics Sonochemistry*, *101*, 106649. https://doi.org/10.1016/j.ultsonch.2023.106649

[8] **Liu, Y.**, Cheng, C., Ziaei-Rad, V., & Shen, Y. (2021). A micromechanics-informed phase field model for brittle fracture accounting for unilateral constraint. *Engineering Fracture Mechanics*, *241*, 107358. https://doi.org/10.1016/j.engfracmech.2020.107358

[9] Chen, C., **Liu, Y.**, He, X., Li, H., Chen, Y., Wei, Y., Zhao, Y., Ma, Y., Chen, Z., Zheng, X., & Liu, H. (2021). Multiresponse shape-memory nanocomposite with a reversible cycle for powerful artificial muscles. *Chemistry of Materials*, *33*(3), 987–997. https://doi.org/10.1021/acs.chemmater.0c04170

[10] **Liu, Y.**, Weng, K., & Shen, Y. (2020). A manifold learning approach to accelerate phase field fracture simulations in the representative volume element. *SN Applied Sciences*, *2*(10), 1682. https://doi.org/10.1007/s42452-020-03468-6

Presentations

2026 | USACM Student Chapter Seminar Series | Online | Seminar presentation
Damage sensing digital twin for piezocomposites integrating neural operator with parametrically upscaled constitutive damage model

2026 | 20th U.S. National Congress on Theoretical and Applied Mechanics (USNCTAM26) | Pasadena, California | Oral presentation
Virtual damage sensing digital twin for piezocomposites integrating neural operator with parametrically upscaled constitutive damage model

2026 | Enabling Multifunctionality through AI Informed Material Discovery and System Design (GRC 2026) | Ventura, California | Poster presentation
Virtual damage sensing digital twin for piezocomposites integrating neural operator with parametrically upscaled constitutive damage model

2026 | HEMI Day 2026 | Baltimore, Maryland | Podium presentation
Virtual damage sensing digital twin for piezocomposites integrating neural operator with parametrically upscaled constitutive damage model

2024 | The 16th World Congress on Computational Mechanics (WCCM16) | Vancouver, Canada
A phase-field approach for the nucleation and propagation of dynamic cracks

2023 | The 17th United States National Congress on Computational Mechanics (USNCCM17) | Albuquerque, New Mexico
A model-based simulation framework for coupled acoustics, elastodynamics, and damage with application to nano-pulse lithotripsy

2022 | The 15th World Congress on Computational Mechanics (WCCM15) | Yokohama, Japan
Model-based simulations of pulsed laser ablation using a CutFEM method

2019 | The 15th United States National Congress on Computational Mechanics (USNCCM15) | Austin, Texas
A homogenization-based phase field approach to fracture

2019 | The 2nd International Conference of Mechanics of Advanced Materials and Structures | Nanjing, China
A micromechanics-based phase field approach to fracture

2019 | The International Conference on Data Driven Computing and Machine Learning in Engineering | Shanghai, China
A manifold learning approach for multiscale phase field evolution for fracture

Journal Referee Service

- Mechanics of Materials
- Advances in Engineering Software
- SoftwareX